

ServiceNow – SAP ECC IDoc

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Scenarios

- In this scenario we will execute several actions from SAP ECC IDoc Spoke including Update Material and Create Purchase Order.

Assumptions

- ServiceNow ECC IDoc Spoke is installed in the ServiceNow instance.
- Infrastructure: ServiceNow Integration Hub, and SAP ECC IDoc are all set-up.

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SAP Configuration (ECC IDoc)

Purpose: In the Tcode BD54, define logical systems

Change View "Logical Systems": Overview

< SAP

Change View "Logical Systems"

✓

New Entries

Cancel

Logical Systems

Log.System

Name

☐

ECXCLNT001

ECC

☐

PODCLNT000

POD CLIENT

→ Position...

Entry 1 of 2

Partner profiles

Partner

Description

▼ Partner Profiles

• Partner Type AD

• Partner Type B

• Partner Type BP

• Partner Type GP

• Partner Type KU

• Partner Type LI

▶ Partner Type LS

• Partner Type US

Bank

Benefits provider

Business Partner

Customer

Vendor

Logical system

User (first 10 characters, no check

Partner No.

PODCLNT00

Partn.Type

LS

Post Processing: Valid Processors

Classification

Telephony

Ty.

A

Work Center

Agent

50000152

Dummy work center

Lang.

EN

English

ServiceNow Integration Hub's SAP ECC IDoc Spoke Configuration

Purpose: in the previous section, the SAP ECC IDoc has been configured so that it can be used with ECC IDoc Spoke actions. All actions are performing change in the SAP system through the java connector.

Step 1 – ServiceNow: Create Credential Object

1. Provide Username and password for ECC IDoc endpoint connection. (this will be provided by SAP team)

The screenshot shows the 'Basic Auth Credentials' configuration page in ServiceNow. The 'Name' field is 'SAP ECC IDOC' and the 'Order' is '100'. The 'User name' field contains 'RFC_USER' and the 'Password' field contains a masked password. The 'Active' checkbox is checked. Below the form is a blue informational bar stating: 'A connection alias resolves your connection and credential at runtime. More than one Credential can be active per Connection Alias at a time. If more than one credential is active, they will be used in order.' Below this is a 'Credential alias' field with a lock icon. At the bottom, there is a table titled 'Discovery IP Affinity' with a search bar and a 'New' button. The table has columns: 'MID server', 'Credential ID', 'IP address', 'IP address decimal', 'Domain', and 'Type'. The table is currently empty, showing 'No records to display'.

MID server	Credential ID	IP address	IP address decimal	Domain	Type
No records to display					

Step 2 – ServiceNow: Create Connection in SAP_IDoc connection alias

1. Go to Connection & Credential Aliases
2. Open alias SAP_IDoc
3. Create new connection as following: choose credentials created in previous step, provide host that serves as endpoint for IDoc connection, check use mid server option.

< ☰

Connection
SAP ECC IDOC Connection



UpdateDelete

↑ ↓

* Name
SAP ECC IDOC Connection

Credential
SAP ECC IDOC

Active ☒

Domain
global

A connection alias resolves your connection and credential at runtime. Only one Connection is active per Connection Alias at a time. [More Info](#)

* Connection alias
sn_sap_ecc_idoc_sp.SAP_IDoc

* Host
ec2-18-135-28-208.eu-west-2.compute.amazonaws.com

Override default port

Use MID server ☒

Advanced MID Server Configuration

Attributes

MID Selection
Specific MID Server

* MID Server
sapmidlnx

UpdateDelete

Step 3 – ServiceNow: Enable Connection (optional)

If there are multiple connections under the same alias, make sure that newly created connection is active.

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Connection & Credential Aliases
sn_sap_ecc_idoc_sp.SAP_IDoc

⚙

⋮

↑

↓

ⓘ

This record is in the [SAP ECC IDoc Spoke](#) application, but [Global](#) is the current application. To edit this record click [here](#).

NameSAP_IDoc

IDsn_sap_ecc_idoc_sp.SAP_IDoc

ApplicationSAP ECC IDoc Spoke

TypeConnection and Credential

Connection typeBasic

Support Multiple Active Connections☐

Default Retry Policy

Configuration Template

Select a template for configuring connection and credential records.

Related Links

Create New Connection & Credential

Connections (1)

Connection Attributes (3)

Child Aliases

≡

🔍

Name

Search

⌕

—

Actions on selected rows...

New

Connection alias = sn_sap_ecc_idoc_sp.SAP_IDoc

☐	🔍	Name ^	Connection type	Active	Credential	Use MID server	Protocol	Host	Override default port	Updated	Updated by	Created	Created by
☐		SAP ECC IDOC Connection	Connection	true	SAP ECC IDOC	true		ec2-18-135-28-208.eu-west-2.compute.amaz...		2023-12-14 16:24:26	admin	2023-12-14 16:24:26	admin

⏪

⏩

1 to 1 of 1

⏪

⏩

Test Case 1: Update Material

ServiceNow will trigger a "update material" request message using Integration Hub's SAP ECC IDoc Spoke. The request message will be passed to the ECC IDoc interface using the java connector. After the response is retrieved, it will be handled back to ServiceNow.

Step 1 – ServiceNow: Configure Action Update Material

1. Create workflow with name e.g. ECC IDOC - PROCUREMENT
2. Add action from ECC IDoc Spoke: Update Material
3. Configure action as following and save it

Input Details

- Control Record
 - Sender Port: 8000
 - Partner Number of Sender: PODCLNT000
- Master Material General Data, Item 1
 - Material: A123
 - Material Group: 01
 - Base Unit of Measure [MEINS]: EA
 - Lab/Office: 001
 - Weight Unit [GEWEI]: KG

ECC IDOC -PROCUREMENT

Inactive

1  Update Material ⓘ

Action

Update Material

* Connection Alias

sn_sap_ecc_idoc_sp.SAP_IDoc

X







▼ * Control Record

Drag and drop object type pill



* Sender Port

8000



* Partner Number of Sender

PODCLNT000



▼ Master Material General Data

Drag and drop array.object type pill





▼ Item 1

Drag and drop object type pill





Function [MSGFN]





Material

A123





Created On





Created By





Last Change





Changed by





Maintenance status





DF at client level





Material Type





Industry Sector





Material Group

01





Old material number [BISMT]





Base Unit of Measure [MEINS]

EA







ECC IDOC -PROCUREMENT

Inactive

Size/dimensions			
Basic material [WRKST]			
Industry Std Desc.			
Lab/Office	001		
Purchasing value key			
Gross Weight			
Net Weight			
Weight Unit [GEWEI]	KG		
Volume			
Volume Unit [VOLEH]			
Container reqmts			
Storage conditions			
Temp. conditions			
Transportation Group			
Haz. material number			
Division			
Competitor			
Number of GR slips			
Procurement rule			
Source of supply			
Season			
Label type			
Label form			

1. Execute workflow

Test Flow

Run your flow to make sure it has no errors before you activate it. When the test finishes running, check the execution details to see each action's configuration, runtime values, and the log messages for any errors that occurred.

This trigger type receives no inputs, test flow execution will run immediately on Test.

☐ Run test in background ⓘ

Cancel
Run Test

Step 2 – ServiceNow: Inspect Request Payload

1. After execution is completed, open execution details
2. Open Step 2 execution details and inspect payload



Step 2 - Ping and Post to SAP [Script]

Integration Metadata

Initial MID Server Selection sapmidlnx

Step Configuration

VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Connection Alias	SAP_IDoc ⓘ	sys_id
Required Runtime	mid	mid
Script	(function execute(inputs, outputs) { try { var credProvFactory = Packages.com.snc.commons.cred...	(function execute(inputs, outputs) { try { var credProvFactory = Packages.com.snc.commons.cred...
Select MID Server Using	use_connection_alias	use_connection_alias
conn_host	ec2-18-135-28-208.eu-west-2.compute.amazonaws.com	conn_host
conn_params	{ "u_client": "000", "u_system_number": "00", "u_lang": "EN" }	conn_params
cred_id	e88d3bea2f0f311047578d1df699b6a6	cred_id
payload	{ "IDOC": { "EDI_DC40": { "SNDPOR": "8000", "SNDPRN": "PODCLNT000", "E1MARAM": { "MFRNR": "", "MSTDE": "", "3r_C_...	json_in

Step 3 – SAP Execution

IDOC is created 198061

IDoc display		Additional information		Short Technical Information	
▼ IDoc 0000000000198061				Direction	2 Inbox
Control Rec.				Current Status	53 OO■
▼ Data records		Total number: 000005		Basic type	MATMAS05
▼ E1MARAM		Segment 000001		Extension	
E1MAKTM		Segment 000002		Message Type	MATMAS
▼ E1MARC		Segment 000003		Partner No.	PODCLNT000
E1MARDM		Segment 000004		Partn.Type	LS
E1MBEWM		Segment 000005		Port	8000
▼ Status records				Content of Selected Segment	
▼ 53		Application document posted		Fld Name	Fld Cont.
Messages have been issued: n					
> 62		IDoc passed to application			
> 64		IDoc ready to be passed to application			
50		IDoc added			

Step 4 – ServiceNow: Inspect Response Payload

1. Open execution details
2. Inspect output data section – it will contain materials response details

1 Update Material

Configuration Details		
VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Connection Alias	SAP_IDoc @	SAP_IDoc @
Control Record	{ "Control Record": { "SNDPOR": "8000", "SNDPRN": "PODCLNT000" } }	
Master Material General Data	{ "Master Material General Data": { "AENAM": null, "AENNR": null, "AESZN": null, "ANP": null, "BEGRU": null, "BE..." } }	

Output Data		
VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Action Status	{ "Action Status": { "code": 0, "message": "Success" } }	
Don't Treat as Error	true	true
Error Message		error_message
Material Response	{ "Material Response": { "docnum": "0000000000198061", "status": "Success", "statusCode": "53", "statusDesc": "... } }	
Status	success	status

Object can be further inspected by clicking on value.

Viewing response [object]



Raw Text

Formatted

```
{
  "Material Response": {
    "docnum": "0000000000198061",
    "status": "Success",
    "statusCode": "53",
    "statusDesc": "Inbound IDoc posted",
    "tid": "0A001E21196F65BBEEF61652"
  }
}
```

Close

Step 5 – Validation on SAP Side

Go to MM04 and give material A123 and observe the changes

Display Changes: Initial Screen

Display Changes: Initial Screen

Selection Parameters

Material	A123
Change Number	
Plant	
Valuation type	
Sales Organization	
Distribution Channel	
Warehouse Number	
Storage Type	
Changed by	
From change date	

Display Changes: Material Overview

< SAP

Display Changes: Material Overview

✓

▼

🔍

List of all change documents for material A123

....

Selected	Date	Time	Changed by	Change no.	Transaction	
☐	21.12.2023	11:04:55	RFC_USER		MM02(ALE)	
☐	21.12.2023	10:08:27	SRAMALINGAM		MM02	

Test Case 2: Create Purchase Order

ServiceNow will trigger an “create purchase order” request message using Integration Hub's SAP ECC IDoc Spoke. The request message will be passed to the ECC IDoc interface using the java connector. After the response is retrieved, it will be handled back to ServiceNow.

Step 1 – ServiceNow: Configure Action Create Purchase Order

1. Create workflow with name e.g. ECC IDOC - PROCUREMENT
2. Add action from ECC IDoc Spoke: Create Purchase Order
3. Configure action as following and save it

Input Details

- Control Record
 - Sender Port: 8000
 - Partner Number of Sender: PODCLNT000
- PO Header
 - Document Date: 20231221
 - Purchasing Doc. Type: NB
 - Purch. Doc. Category: F
 - Company Code: 9000
 - Purch. Organization: 9000
 - Purchasing Group: 009
 - Vendor: A12346
- PO Lines > Line 1
 - Item [PO_ITEM]: 00010
 - Material [PUR_MAT]: A123
 - Storage Location: ROH1
 - Material Group: 01
 - Short Text: IC ENGINE
 - Plant: 9000
 - Order Unit: EA
 - Net order price: 100
 - Price unit: 1
 - Order Price Unit: EA
 - Tax Code: V0
 - Line Items Delivery Schedule > Delivery Schedule 1
 - Item: 00010
 - Delivery date: 20231221
 - Scheduled Quantity: 100
- Addresses for Inbound Delivery > Address 1
 - Item: 00010

- Name: GL PLANT
- City: BANGALORE
- Postal Code: 638001

 **ECC IDOC -PROCUREMENT** Inactive

2

 **Create Purchase Order** 

Action

Create Purchase Order

* Connection Alias

sn_sap_ecc_idoc_sp.SAP_IDoc

X







▼ * Control Record

Drag and drop object type pill



* Sender port

8000



* Partner number of sender

PODCLNT000



▼ PO Header

Drag and drop object type pill



Document Date

20231221



* Purchasing Doc. Type

NB



Purch. Doc. Category

F



* Company Code

9000



* Purch. Organization

9000



* Purchasing Group

009



Outline agreement



* Vendor

A12346



Purchasing Document



Supplying Plant



Created By



Language Key



ISO code for language key



▼ * PO Lines

Drag and drop array.object type pill







ECC IDOC -PROCUREMENT

Inactive

▼ * PO Lines

Drag and drop array.object type pill



▼ Line 1

Drag and drop object type pill



Purchasing Document			
Item [PO_ITEM]	00010		
Address			
Material [MATERIAL]			
Material [PUR_MAT]	A123		
Purchasing info rec.			
Item category			
Acct Assignment Cat.			
Outline agreement			
Princ.agreement item			
Storage Location	ROH1		
Material Group	01		
Short Text	IC ENGINE		
Distribut. indicator			
Partial invoice			
Kanban Indicator			
Plant	9000		
Allocation Table			
Item [AT_ITEM]			
Order Unit	EA		
Net order price	100		



ECC IDOC -PROCUREMENT

Inactive

Allocation Table			
Item [AT_ITEM]			
Order Unit	EA		
Net order price	100		
Price unit	1		
Quantity Conversion [CONV_NUM1]			
Quantity Conversion [CONV_DEN1]			
Order Price Unit	EA		
Package number			
Promotion			
Acknowledgment Reqd			
Req. Tracking Number			
Planned Deliv. Time			
Returns Item			
Alloc.-Tab.-Relevant			
Vendor material no.			
Mfr part profile			
Manufacturer Part No.			
Manufacturer			
Ext. manufacturer			
Adopt price			
Shipping Instr.			
Item Category			

Delivery Completed			
Deletion indicator			
No Rounding			
Tax Code	V0		
Material Number [MATERIAL_EXTERNAL]			
GUID (External) [MATERIAL_GUID]			



ECC IDOC -PROCUREMENT

Inactive

▼ Addresses for
Inbound
Delivery

Drag and drop array.object type pill



▼ Address 1

Drag and drop object type pill



Item	00010		
Address number			
Form of address			
Name	GL PLANT		
Name 2			
Name 3			
Name 4			
c/o			
City	BANGALORE		
District			
City Code [CITY_NO]			
Postal Code	638001		
PO Box Postal Code			
Company Postal Code			
PO Box			
PO Box City			
Delivery District			
Street			
Street Code			
Street Abbreviation			
House Number			

1. Execute workflow

Test Flow

Run your flow to make sure it has no errors before you activate it. When the test finishes running, check the execution details to see each action's configuration, runtime values, and the log messages for any errors that occurred.

This trigger type receives no inputs, test flow execution will run immediately on Test.

☐ Run test in background ⓘ

Cancel
Run Test

Step 2 – ServiceNow: Inspect Request Payload

1. After execution is completed, open execution details
2. Open Step 2 execution details and inspect payload

 Step 2 - Ping and Post to SAP [Script]

Integration Metadata

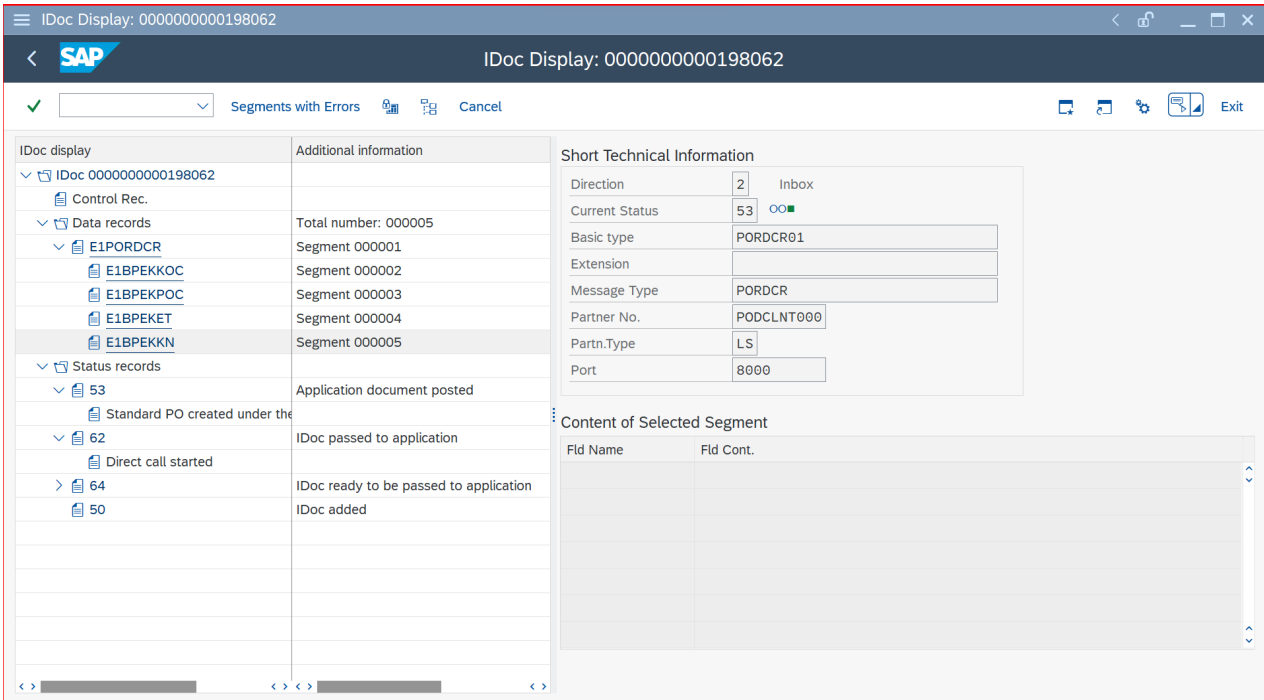
Initial MID Server Selection [sapmidlnx](#)

Step Configuration

VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Connection Alias	SAP_IDoc ⓘ	sys_id ⓘ
Required Runtime	mid	mid
Script	<pre>(function execute(inputs, outputs) { try { var credProvFactory = Packages.com.snc.commons.cred...</pre>	<pre>(function execute(inputs, outputs) { try { var credProvFactory = Packages.com</pre>
Select MID Server Using	use_connection_alias	use_connection_alias
conn_host	ec2-18-135-28-208.eu-west-2.compute.amazonaws.com	conn_host ⓘ
conn_params	{ "u_client": "000", "u_system_number": "00", "u_lang": "EN" }	conn_params ⓘ
cred_id	e88d3bea2f0f311047578d1df699b6a6	cred_id ⓘ
payload	<pre>{ "IDOC": { "EDI_DC40": { "SNDPOR": "8000", "SNDPRN": "PODCLNT000", "E1PORDCR": {"E1BPEKKOC": {"CREATED_BY": "" ...</pre>	json_in ⓘ

Step 3 – SAP Execution

The IDOC "198062" was successfully received and PO was created.



Step 4 – ServiceNow: Inspect Response Payload

1. Open execution details
2. Inspect output data section – it will contain materials response details

Output Data		
VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Action Status	{ "Action Status": { "code": 0, "message": "Success" } }	
Don't Treat as Error	true	true
Error Message		error_message
Purchase Order Response	{ "Purchase Order Response": { "docnum": "0000000000198062", "status": "Success", "statusCode": "53", "status..." } }	
Status	success	status

Object can be further inspected by clicking on value.

Viewing response [object]



Raw TextFormatted

```
{
  "Purchase Order Response": {
    "docnum": "0000000000198062",
    "status": "Success",
    "statusCode": "53",
    "statusDesc": "Inbound IDoc posted",
    "tid": "0A001E21196F65BBEEFE1653"
  }
}
```

Close

Step 5 – Validation on SAP Side

<

SAP

✓

▼

Segments with Errors

IDoc display

▼

 IDoc 0000000000198062

Control Rec.

▼

 Data records

▼

 E1PORDCR

E1BPEKKOC

E1BPEKPOC

E1BPEKET

E1BPEKKN

▼

 Status records

▼

 53

Standard PO created under the number 4500000046

▼

 62

Direct call started

>

 64

50

Standard PO 4500000046 Created by RFC

SAP Standard PO 4500000046 Created by RFC

Document Overview On Print Preview Messages More

	S...	Itm	A	I	Material	Short Text	PO Quantity	O... C Deliv. Date	Net Price	Curre...	Per	O...	Matl Group	PInt
☐	☐	10			A123	IC ENGINE		100EA D 21.12.2023		100,00 INR	1	EA	Material grou...	GL plant

Add Planning

[10] A123 , IC ENGINE